# Supplementary Material for End-to-End Self-Supervised Optimization for Stochastic Distribution Network Reconfiguration

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This supplement contains the operating model, supporting derivations and proofs, and implementation and evaluation details cited in Sections II–IV of the Letter. Notation follows the Letter; references are numbered independently.

## S1. Complete Operating Model

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite connected candidate graph with $n$ buses and $m$ branches, and let $\mathcal{T}$ be its nonempty set of spanning trees. Binary $z_e$ indicates whether branch $e$ is closed, and $z^0 \in \mathcal{T}$ is the fixed initial topology. A policy selects $z \in \mathcal{T}$ from predecision information $x$ and holds it for 24 h. Continuous dispatch adapts to the realized daily load–PV trajectory $\xi$. Orient the selected tree away from source bus 0 and denote its directed branches by $\mathcal{E}_z$.

### *S1.1. Power Balances and Network Constraints*

All power variables are in per unit on base $S_{\text{base}}$ (MVA), with hourly intervals $\Delta t = 1$ h. Let $p^d_{j,t}, q^d_{j,t}$ denote load and $\bar{p}^{\text{pv}}_{j,t}$ available photovoltaic (PV) generation. The controls $p^{\text{shed}}_{j,t}$ and $p^{\text{pv}}_{j,t}$ are shed active load and utilized PV generation, respectively. PV operates at unity power factor. With $\kappa_j = q^d_{j,t}/p^d_{j,t}$ fixed by the load power factor, net demands are

$$\begin{aligned} p^{\text{net}}_{j,t} &= p^d_{j,t} - p^{\text{shed}}_{j,t} - p^{\text{pv}}_{j,t}, \\ q^{\text{net}}_{j,t} &= q^d_{j,t} - \kappa_j p^{\text{shed}}_{j,t}. \end{aligned} \tag{S1}$$

For zero-load buses, $\kappa_j$ is set to zero. Load and PV controls satisfy

$$\begin{aligned} 0 \le p^{\text{shed}}_{j,t} \le p^d_{j,t}, \\ 0 \le p^{\text{pv}}_{j,t} \le \bar{p}^{\text{pv}}_{j,t}. \end{aligned} \tag{S2}$$

For each closed branch $(i,j) \in \mathcal{E}_z$, the branch-flow constraints are

$$\begin{aligned} P_{ij,t} &= \sum_{k:(j,k)\in\mathcal{E}_z} P_{jk,t} + p^{\text{net}}_{j,t} + r_{ij}\ell_{ij,t}, \\ Q_{ij,t} &= \sum_{k:(j,k)\in\mathcal{E}_z} Q_{jk,t} + q^{\text{net}}_{j,t} + x_{ij}\ell_{ij,t}, \\ v_{j,t} &= v_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) \\ &\quad + (r^2_{ij} + x^2_{ij})\ell_{ij,t}, \\ P^2_{ij,t} + Q^2_{ij,t} &\le v_{i,t}\ell_{ij,t}. \end{aligned} \tag{S3}$$

Here $P_{ij,t}, Q_{ij,t}$ are sending-end active and reactive power; $r_{ij}, x_{ij}$ are per-unit resistance and reactance. The variables $v_{j,t}$ and $\ell_{ij,t}$ are squared voltage and current magnitudes. Branch reactance $x_{ij}$ is distinct from the policy input $x$. The last inequality relaxes the branch-flow current–power equality [1].

The source balances and fixed source voltage are

$$\begin{aligned} P_{0,t} &= \sum_{k:(0,k)\in\mathcal{E}_z} P_{0k,t} + p^{\text{net}}_{0,t}, \\ Q_{0,t} &= \sum_{k:(0,k)\in\mathcal{E}_z} Q_{0k,t} + q^{\text{net}}_{0,t}, \\ v_{0,t} &= v_0. \end{aligned} \tag{S4}$$

Network and source limits are

$$\begin{aligned} \underline{V}^2 \le v_{j,t} \le \overline{V}^2, \quad 0 \le \ell_{ij,t} \le \overline{I}^2_{ij}, \\ P^2_{ij,t} + Q^2_{ij,t} \le \overline{S}^2_{ij}, \\ (P_{ij,t} - r_{ij}\ell_{ij,t})^2 + (Q_{ij,t} - x_{ij}\ell_{ij,t})^2 \\ \le \overline{S}^2_{ij}, \\ -\overline{S}_0(z) \le P_{0,t} \le \overline{S}_0(z), \quad 0 \le Q_{0,t} \le \overline{S}_0(z). \end{aligned} \tag{S5}$$

The experiments use $\underline{V} = 0.95$, $\overline{V} = 1.05$, $v_0 = 1$, and $\overline{I}_{ij} = \overline{S}_{ij}/\underline{V}$. The source limit is $\overline{S}_0(z) = \sum_{k:(0,k)\in\mathcal{E}_z} \overline{S}_{0k}$; active export is allowed and reactive export is excluded. The maximum over candidate trees bounds root import uniformly. Open branches carry zero power and current and impose no voltage-drop equation.

### *S1.2. Dispatch and Total Cost*

Daily energies in MWh are

$$\begin{aligned} E_\ell &= S_{\text{base}}\Delta t \sum_t \sum_{(i,j)\in\mathcal{E}_z} r_{ij}\ell_{ij,t}, \\ E_{\text{curt}} &= S_{\text{base}}\Delta t \sum_t \sum_{j\in\mathcal{V}} (\bar{p}^{\text{pv}}_{j,t} - p^{\text{pv}}_{j,t}), \\ E_{\text{shed}} &= S_{\text{base}}\Delta t \sum_t \sum_{j\in\mathcal{V}} p^{\text{shed}}_{j,t}. \end{aligned} \tag{S6}$$

The set $\mathcal{U}_{\text{soc}}(z,\xi)$ consists of all continuous variables satisfying (S1)–(S5) for the trajectory $\xi$. The recourse and total costs are

$$\begin{aligned} Q(z,\xi) &= \min_{u\in\mathcal{U}_{\text{soc}}(z,\xi)} \big(c_\ell E_\ell + c_{\text{pv}} E_{\text{curt}} + c_s E_{\text{shed}}\big), \\ C(z,\xi) &= c_{\text{sw}}\|z - z^0\|_1 + Q(z,\xi). \end{aligned} \tag{S7}$$

Here, $s(z) = c_{\text{sw}}\|z - z^0\|_1$ is the switching cost. All coefficients are nonnegative. Reported operating cost includes losses, curtailment, and switching, but excludes the shedding

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penalty; the optimization objective includes all four components. No electricity purchase or sale settlement is modeled.

Full shedding removes active and reactive demand, and full PV curtailment is permitted. Radiality alone therefore does not ensure that all demand is served. Throughout the proofs, $C$ denotes the exact second-order-cone programming (SOCP) optimal value. The exact radial AC model replaces the last inequality in (S3) with equality; the theoretical bound does not assert a corresponding AC guarantee.

## S2. Supporting Derivations and Proofs

### *S2.1. Self-Supervised Training Identities*

For branch scores $s_\theta(x)$ and temperature $\tau > 0$, weighted Wilson sampling with conductances $w_e = \exp(s_e/\tau)$ draws trees from

$$\pi_\theta(z \mid x) = \frac{\exp(s_\theta(x)^\top z/\tau)}{\sum_{t\in\mathcal{T}} \exp(s_\theta(x)^\top t/\tau)}. \tag{S8}$$

The tree weight is proportional to $\prod_{e\in z} w_e$ [2]. The following derivation assumes exact sampling without walk truncation.

Fix an update's history, input $x$, future $\xi$, and feedback function

$$L(z) = C(z,\xi)/a + \lambda \mathbf{1}\{E_{\mathrm{shed}}(z,\xi) > \varepsilon_{\mathrm{shed}}\},$$

where $a > 0$ and $\lambda$ are held fixed. Let $J_s$ be the Jacobian of $s_\theta(x)$. At differentiable network parameters, differentiating the finite sum gives

$$\begin{aligned} \nabla_\theta \mathbb{E}_\pi L(Z) &= \mathbb{E}_\pi[L(Z)\nabla_\theta \log \pi_\theta(Z \mid x)], \\ \nabla_\theta \log \pi_\theta(z \mid x) &= \tau^{-1} J_s^\top (z - \mathbb{E}_\pi Z). \end{aligned} \tag{S9}$$

For $J \geq 2$ independent tree draws, set $L_r = L(Z_r)$, $b_{-r} = (J-1)^{-1}\sum_{q\neq r} L_q$, and $A_r = L_r - b_{-r}$. Independence of $b_{-r}$ and $Z_r$, together with $\mathbb{E}_\pi \nabla \log \pi_\theta(Z \mid x) = 0$, shows that

$$\begin{aligned} \widehat{d} &= \frac{1}{J}\sum_r A_r \nabla_\theta \log \pi_\theta(Z_r \mid x) \\ &= \frac{1}{J\tau}\sum_r A_r J_s^\top Z_r \end{aligned} \tag{S10}$$

is unbiased for the expected-feedback gradient in (S9). The second equality follows from $\sum_r A_r = 0$. Holding the sampled trees and feedback fixed therefore makes the gradient of the Letter's surrogate equal to $\widehat{d}$, without differentiating through either the dispatch solver or the tree decoder.

The implementation's equivalent pairwise surrogate is

$$\begin{aligned} \mathcal{L}_{\mathrm{PG}} = \frac{1}{J(J-1)\tau}\sum_{r<q} &\mathrm{stopgrad}(L_r - L_q) \\ &\times s_\theta(x)^\top \mathrm{stopgrad}(Z_r - Z_q), \end{aligned} \tag{S11}$$

where $\mathrm{stopgrad}$ holds its argument fixed during differentiation. Its equivalence follows from

$$\begin{aligned} &\sum_{r<q}(L_r - L_q)(Z_r - Z_q) \\ &\quad = J\sum_r L_r Z_r - \Big(\sum_r L_r\Big)\Big(\sum_r Z_r\Big). \end{aligned}$$

For the structured hinge in the Letter, $d(z,z^+) = \sum_e z_e(1 - z_e^+)$ because every tree has $n-1$ branches. Its maximizing tree is thus a maximum spanning tree (MST) with modified scores $s_e + \mu(1 - z_e^+)$.

### *S2.2. Expected Optimality Gap and Conditional Asymptotic Optimality*

Let $\mathcal{H}$ be the class of all measurable policies from the input space $\mathcal{X}$ to $\mathcal{T}$, and let $\mathcal{H}_N \subseteq \mathcal{H}$ be the neural policy class with deterministic MST decoding and fixed tie breaking. Define

$$\begin{aligned} R(g) &= \mathbb{E}[C(g(X),\Xi)], \\ R^* &= \inf_{g\in\mathcal{H}} R(g), \\ \varepsilon_{\mathrm{app}}(N) &= \inf_{g\in\mathcal{H}_N} R(g) - R^*. \end{aligned} \tag{S12}$$

Since $\mathcal{T}$ is finite, measurable conditional costs admit a measurable optimal tree selector by fixed tie breaking. Hence $R^*$ is the minimum expected cost attainable using the same predecision information $X$.

*1) Assumptions:* The graph, prices, capacities, feature map, and class $\mathcal{H}_N$ are fixed before sampling. Losses, conditional costs, and the suprema below are measurable, or admit separable measurable versions. The $N$ training observations are independent and identically distributed groups $Z_i = (X_i, \Xi_{i1}, \ldots, \Xi_{iK})$. Each $\Xi_{ik} \mid X_i$ has the target conditional future distribution; dependence within a group is allowed. The baseline $B$ defined below is integrable.

Full active-load shedding removes the corresponding reactive demand, and full PV curtailment is allowed. These actions admit zero root injection, zero branch flows and currents, and a common permissible voltage. The exact recourse optimum in (S7) is finite and attained. Daily net root import and available PV energy are bounded above by $\bar{E}_0$ and $\bar{G}$, uniformly over candidate trees; losses and all prices are nonnegative. For $n$ buses and $m$ candidate branches, switching satisfies

$$0 \leq s(z) \leq \bar{S} := 2c_{\mathrm{sw}} \min\{n-1, m-n+1\}.$$

This follows because two spanning trees exchange at most $\min\{n-1, m-n+1\}$ branches. A common root-power bound $\bar{P}_0$ in MW gives $\bar{E}_0 = 24\bar{P}_0$ in MWh; PV capacities and bounded availability give $\bar{G}$.

For total demand $D(\xi)$ and available PV energy $G(\xi)$, set

$$\begin{aligned} B(\xi) &= c_s D(\xi) + c_{\mathrm{pv}} G(\xi), \\ c(z,\xi) &= C(z,\xi) - B(\xi), \\ A &= c_s(\bar{E}_0 + \bar{G}) + c_{\mathrm{pv}}\bar{G}, \\ W &= A + \bar{S}. \end{aligned} \tag{S13}$$

**Lemma S2.1** (Bounded centered cost)**:** *Under the stated physical assumptions, $-A \leq c(z,\xi) \leq \bar{S}$ for every tree and trajectory. Consequently, $|C(z,\xi) - C(t,\xi)| \leq W$ for any two trees.*

*Proof:* The full-shedding/full-curtailment dispatch costs $B(\xi) + s(z)$, giving the upper bound. Summed active-power balances give $D_{\mathrm{served}} = E_{\mathrm{root}} + G_{\mathrm{used}} - E_\ell \leq \bar{E}_0 + \bar{G}$. Every feasible dispatch's total cost minus $B$ is therefore

$$c_\ell E_\ell - c_s D_{\mathrm{served}} - c_{\mathrm{pv}} G_{\mathrm{used}} + s(z) \geq -A.$$

Taking the minimum over dispatches preserves this lower bound. The centered interval has width $W$. ∎

No bounded support for demand is required: the potentially unbounded baseline is common to all policies.

Define the empirical total cost and centered group losses by

$$\begin{aligned} \widehat{R}_N(g) &= \frac{1}{NK}\sum_{i,k} C(g(X_i), \Xi_{ik}), \\ f_g(Z) &= \frac{1}{K}\sum_{k} c(g(X), \Xi_k), \\ \mathcal{F}_N &= \{f_g : g \in \mathcal{H}_N\}. \end{aligned} \tag{S14}$$

Write $R_c(g) = \mathbb{E} f_g(Z)$ and $\widehat{R}_{c,N}(g) = N^{-1}\sum_i f_g(Z_i)$. Then

$$\begin{aligned} R_c(g) &= R(g) - \mathbb{E}B(\Xi), \\ \widehat{R}_{c,N}(g) &= \widehat{R}_N(g) - \widehat{B}_N, \\ \widehat{B}_N &= (NK)^{-1}\sum_{i,k} B(\Xi_{ik}). \end{aligned}$$

Centering therefore preserves risk differences and empirical minimizers. For independent Rademacher signs $\sigma_i$, use

$$\mathfrak{R}_N(\mathcal{F}_N) = \mathbb{E}_{Z_1,\dots,Z_N,\sigma} \sup_{g\in\mathcal{H}_N} \frac{1}{N}\sum_i \sigma_i f_g(Z_i).$$

**Lemma S2.2** (Uniform generalization)**:** *For $0 < \delta < 1$, with probability at least $1-\delta$, simultaneously for all $g \in \mathcal{H}_N$,*

$$\begin{aligned} |R_c(g) - \widehat{R}_{c,N}(g)| &\le e_N(\delta), \\ e_N(\delta) &:= 2\mathfrak{R}_N(\mathcal{F}_N) \\ &\quad + W\sqrt{\frac{\log(2/\delta)}{2N}}. \end{aligned} \tag{S15}$$

*Proof:* Let $\Phi_+ = \sup_g(R_c(g) - \widehat{R}_{c,N}(g))$ and $\Phi_- = \sup_g(\widehat{R}_{c,N}(g) - R_c(g))$. With independent copies $Z_i'$, the standard symmetrization argument [3] gives

$$\begin{aligned} \mathbb{E}\Phi_+ &\le \mathbb{E}\sup_g \frac{1}{N}\sum_i [f_g(Z_i') - f_g(Z_i)] \\ &= \mathbb{E}\sup_g \frac{1}{N}\sum_i \sigma_i [f_g(Z_i') - f_g(Z_i)] \\ &\le 2\mathfrak{R}_N(\mathcal{F}_N). \end{aligned}$$

The same bound holds for $\mathbb{E}\Phi_-$ by reversing the difference. Replacing one independent group changes either supremum by at most $W/N$. The bounded-difference inequality gives $\mathbb{P}(\Phi_\pm - \mathbb{E}\Phi_\pm > t) \le \exp(-2Nt^2/W^2)$. Set $t = W\sqrt{\log(2/\delta)/(2N)}$ and apply a union bound to the two tails. ∎

The independent sample count is $N$, not $NK$; within-group dependence does not affect this argument.

Let $\widetilde{R}_N$ denote numerical empirical total cost and assume

$$\sup_{g\in\mathcal{H}_N} |\widetilde{R}_N(g) - \widehat{R}_N(g)| \le \eta_N.$$

For the returned policy $\hat{g}_N \in \mathcal{H}_N$, define its numerical empirical optimization residual

$$\varepsilon_{\mathrm{alg}}(N) = \widetilde{R}_N(\hat{g}_N) - \inf_{g\in\mathcal{H}_N} \widetilde{R}_N(g) \ge 0. \tag{S16}$$

This residual includes differences between the training losses and total cost; it need not be known or small. The numerical bound is assumed to hold almost surely. If it holds only on an additional event, that event's failure probability must be added to the bound below.

**Theorem S2.3** (Expected optimality gap)**:** *Under the stated assumptions, with probability at least $1-\delta$ over the training groups,*

$$\begin{aligned} G_N &:= R(\hat{g}_N) - R^* \\ &\le 4\mathfrak{R}_N(\mathcal{F}_N) + W\sqrt{\frac{2\log(2/\delta)}{N}} \\ &\quad + \varepsilon_{\mathrm{app}}(N) + \varepsilon_{\mathrm{alg}}(N) + 2\eta_N. \end{aligned} \tag{S17}$$

*Moreover, $0 \le G_N \le W$.*

*Proof:* On the event in Lemma S2.2, choose, for any $\zeta > 0$, $g_\zeta \in \mathcal{H}_N$ with $R(g_\zeta) \le \inf_{g\in\mathcal{H}_N} R(g) + \zeta$. Set $\widetilde{R}_{c,N} = \widetilde{R}_N - \widehat{B}_N$. Uniform generalization and numerical accuracy give

$$\begin{aligned} R_c(\hat{g}_N) &\le \widehat{R}_{c,N}(\hat{g}_N) + e_N \\ &\le \widetilde{R}_{c,N}(\hat{g}_N) + \eta_N + e_N \\ &\le \widetilde{R}_{c,N}(g_\zeta) + \varepsilon_{\mathrm{alg}}(N) + \eta_N + e_N \\ &\le \widehat{R}_{c,N}(g_\zeta) + \varepsilon_{\mathrm{alg}}(N) + 2\eta_N + e_N \\ &\le R_c(g_\zeta) + \varepsilon_{\mathrm{alg}}(N) + 2\eta_N + 2e_N. \end{aligned}$$

Adding the common baseline expectation, subtracting $R^*$, and letting $\zeta \downarrow 0$ proves (S17). Lemma S2.1 also bounds conditional regret relative to the optimal tree by $W$; averaging proves $0 \le G_N \le W$. ∎

**Corollary S2.4** (Conditional asymptotic optimality)**:** *For a predetermined sequence $\mathcal{H}_N$ with fixed physical width $W$, suppose $\mathfrak{R}_N(\mathcal{F}_N) \to 0$ and $\varepsilon_{\mathrm{app}}(N) \to 0$. If $\varepsilon_{\mathrm{alg}}(N), \eta_N \to 0$ in probability, then $G_N \to 0$ in probability and in mean over training and algorithm randomness.*

*Proof:* Take $\delta_N = N^{-2}$ for $N \ge 2$ in (S17). Its deterministic statistical terms vanish and its remaining terms tend to zero in probability. For any $t > 0$, $\mathbb{P}(G_N > t)$ is at most $\delta_N$ plus the probability that the displayed upper bound exceeds $t$, and thus tends to zero. Since $0 \le G_N \le W$, $\mathbb{E}G_N \le t + W\mathbb{P}(G_N > t)$; let $N \to \infty$ and then $t \downarrow 0$. ∎

The expectation in $R$ concerns future operation, while (S17) is a high-probability statement over training data. These results concern exact SOCP total costs and the returned deterministic policy, including any checkpoint selection. They do not assert AC exactness or that the performed experiments establish vanishing optimization and numerical residuals. Increasing the sample count alone does not establish all conditions of Corollary S2.4.

## S3. Implementation and Evaluation Details

### *S3.1. Systems and Data*

The main comparison uses adapted balanced 33-, 69-, 136-, and 415-bus feeders, with uniform branch thermal ratings of 6.5540, 6.9903, 29.9370, and 45.6074 MVA, respectively. All methods share each feeder's physical settings and test trajectories. Each system has 128 training inputs with four

TABLE S1
TRAINING SETTINGS FOR THE MAIN COMPARISON

| Setting | Value |
|---|---|
| Hidden layers | Three ReLU layers, width 128 each |
| Mini-batch size; epochs | 4; 100 |
| Optimizer; weight decay | AdamW; $10^{-4}$ |
| Learning rate; training seed | 0.003; 2203 |
| Gradient-norm clipping | 10 |
| Sampled trees; temperature | $J = 4$; $\tau = 0.5$ |
| Refinement weight; margin | $\beta = 1$; $\mu = 0.1$ |
| Local branch-exchange candidates | 2 per input |
| Shedding-event threshold | $\varepsilon_{\mathrm{shed}} = 10^{-6}$ MWh |
| Calibration trees per input | $J_{\mathrm{cal}} = 4$ |
| Risk-weight initial value; range | $\lambda_1 = 0$; $[0, 10]$ |
| Risk-update step; reference rate | 0.1; 0.01 |

daily futures per input. Independent validation sets contain 32, 16, 32, and 32 inputs, respectively, with 16 futures per input. The shared test set contains 64 inputs, each with 16 futures. A method fixes one topology per input, and each plotted observation averages that input's futures.

The 15 predecision features describe system load, four load-zone multipliers, morning and evening peaks, solar scale, cloud level, solar shift, and five uncertainty parameters. The latter specify load and PV variability, spatial and temporal correlation, and shared load–PV correlation. Fixed affine normalization precedes the score network. Raw network outputs $h_e$ are transformed as

$$\begin{aligned} \widetilde{s}_e &= b \tanh(h_e/b), \qquad b = 2, \\ s_e &= \widetilde{s}_e - \frac{1}{m}\sum_f \widetilde{s}_f. \end{aligned} \tag{S18}$$

This transformation preserves branch rankings and ties in exact arithmetic. Deployment uses Kruskal's MST decoder with fixed branch-ID tie breaking.

### *S3.2. Policy Training*

Table S1 lists the main training settings. An epoch is one pass through the training inputs. Policy-gradient updates use one trajectory per input, cycling through the four trajectories across epochs. Refinement evaluates each candidate on all four training trajectories. Training and validation dispatch use Gurobi for the 33- and 69-bus systems and Clarabel for the 136- and 415-bus systems. Mean validation total cost selects the proposed method's checkpoint. Validation includes epoch 0 and every second epoch for 33 buses or every fifth epoch for the other systems; ties retain the earlier checkpoint.

Before training, the initial policy independently samples $J_{\mathrm{cal}} = 4$ trees per training input, with replacement. Each tree is evaluated on all $K = 4$ training futures, retaining sampling multiplicities. With costs expressed in USD, the fixed normalization is

$$\begin{aligned} \overline{C}_0 &= \frac{1}{NJ_{\mathrm{cal}}K}\sum_{i=1}^{N}\sum_{r=1}^{J_{\mathrm{cal}}}\sum_{k=1}^{K} C(z_{ir}^{\mathrm{cal}}, \xi_{ik}), \\ a &= \max\{862 \text{ USD}, \overline{C}_0\}, \end{aligned} \tag{S19}$$

where $z_{ir}^{\mathrm{cal}}$ is sampled from the initial policy. Calibration uses an independent random stream; these trees do not initialize the candidate archive or enter the policy-gradient samples.

The risk weight is fixed within each epoch. After epoch $e$, it is updated as

$$\lambda_{e+1} = \min\{10, \max\{0, \lambda_e + 0.1(\widehat{p}_e - 0.01)\}\}, \tag{S20}$$

where $\widehat{p}_e$ is the observed shedding-event frequency among all $NJ$ policy-gradient samples in that epoch, retaining sampling multiplicities. The 1% reference frequency guides penalty adaptation; it is not an imposed chance constraint.

Refinement pools the current MST, stored tree, sampled trees, and two local branch-exchange candidates, then removes duplicates. Let $z^{\mathrm{cur}}$ be the current MST and let $\overline{C}(z)$, $\overline{p}(z)$, and $\overline{E}(z)$ denote mean total cost, shedding-event frequency, and mean shed energy over the same four training futures. A candidate is admissible when

$$\begin{aligned} &\overline{C}(z) < \overline{C}(z^{\mathrm{cur}}) - \delta_C, \\ &\overline{p}(z) \le \overline{p}(z^{\mathrm{cur}}) + 10^{-12}, \\ &\overline{E}(z) \le \overline{E}(z^{\mathrm{cur}}) + 10^{-8} \text{ MWh}, \end{aligned} \tag{S21}$$

where $\delta_C = 8.62 \times 10^{-7}$ USD. The admissible candidate with minimum mean total cost defines the hinge target. The stored tree is updated by minimum mean total cost in the candidate pool; it need not itself pass the risk screen. The hinge terms are summed over accepted targets and divided by the full mini-batch size, as in the Letter. Inputs without a target contribute only to the policy-gradient term.

### *S3.3. Benchmarks*

PTO feeds neural point forecasts into a mixed-integer SOCP with a 1000-s time limit and a requested relative MIP gap of 1%. A forecast-based nominal-tree dispatch supplies a warm start independently of the proposed method, with topology variables left free. The requested gap need not be achieved within the time limit.

SL+MST shares the proposed method's architecture and training-sample count. Independent branch-exchange search generates labels using eight rounds and a 32-neighbor budget. Binary cross-entropy trains branch logits, followed by MST decoding. The 33- and 69-bus checkpoints use epoch 100; the 136- and 415-bus checkpoints retain validation selection under the original branch-threshold decoder. Adding MST does not retrain or reselect these checkpoints. The labels are finite-budget decisions without global-optimality certificates.

### *S3.4. Costs, Timing, and Physical Evaluation*

Table S2 reports coefficients after multiplying the implementation's original cost coefficients by 0.862 to express them in USD. The same conversion gives the normalization floor and cost-improvement tolerance in (S19) and (S21). Daily total cost equals reported operating cost plus $8620E_{\mathrm{shed}}$, with $E_{\mathrm{shed}}$ in MWh.

Boxes show medians and interquartile ranges; whiskers extend to observations within 1.5 interquartile ranges. Cost outliers are hidden visually but retained in all statistics; shedding outliers are shown.

TABLE S2
ECONOMIC COEFFICIENTS

| Component and coefficient | Value | Unit |
|---|---|---|
| Energy loss, $c_\ell$ | 86.2 | USD/MWh |
| PV curtailment, $c_{\mathrm{pv}}$ | 43.1 | USD/MWh |
| Unserved energy, $c_s$ | 8620 | USD/MWh |
| Switching, $c_{\mathrm{sw}}$ | 0.2155 | USD per switch-state change |

The main comparison uses an Intel Core Ultra 7 265K CPU, with one thread per physical solve and one CPU thread for neural timing. PTO uses Gurobi 12.0.3. Common physical evaluation uses Gurobi SOCP dispatch with a Clarabel fallback, followed by independent AC checks and, when needed, recovery and rechecking of failed hours. Learned-policy timing includes normalization, a forward pass, and MST decoding, excluding model loading. PTO timing includes initialization, model construction, mixed-integer search, and decision extraction, so wall time can exceed its solver limit. All times exclude subsequent continuous dispatch and AC checks and are averaged over 64 test inputs. Sampled-tree evaluations and refinement occur offline.

Physical assessment for the main comparison allows shedding and independently checks power balances, voltage drops, network limits, and control bounds. Inequality acceptance uses $10^{-4}$, with stricter balance and control checks. This feasibility tolerance is not the numerical cost-error bound $\eta_N$ in the theorem.

### *S3.5. Training-Sample-Size Evaluation Protocol*

The 136-bus study uses four previously trained models with $N = 32, 64, 128$, and $256$, keeping the architecture, mini-batch size, training seed, and 100-epoch schedule fixed. This gives 800, 1600, 3200, and 6400 updates; the $N = 128$ model is reused from the main comparison. Thus, increasing $N$ also increases training effort.

For Fig. 3 of the Letter, the four checkpoints are frozen and evaluated on a common set of 1,000 test inputs, each with 16 daily futures from the original data-generating distribution. An independent random stream with seed 2203 generates this set, with no exact input or daily-trajectory duplicates in the training, validation, or earlier test sets. No retraining or checkpoint reselection uses these results. Each input's total cost is averaged over its 16 futures. The 67 displayed inputs satisfy

$$\overline{C}_{32}(x) > \overline{C}_{64}(x) > \overline{C}_{128}(x) > \overline{C}_{256}(x), \tag{S22}$$

where $\overline{C}_N$ includes the shedding penalty. The figure displays operating cost excluding that penalty and load shedding for this outcome-selected subset. It is descriptive and does not establish generalization or asymptotic optimality; no input-wise optimum or optimality gap is computed.

This separate evaluation uses the original Clarabel SOCP evaluator and its numerical settings. All 24 hourly solves must be accepted for a daily evaluation. It provides numerical SOCP costs without additional AC-feasibility certification for this test set.

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